Date: COAL Assignment #2 Day:

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Q1)
```
INCLUDE Irvine32.inc
.data
.code
main PROC
mov ax, 3
mov bx, ax
mov cx, ax
shl cx, 4
shl bx, 3
shl ax, 1
add ax, bx
add ax, cx
movzx eax, ax
call WriteDec
exit
main ENDP
END main
```


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Q22)

```
INCLUDE Irvine32.inc
.data
num DWORD 2345ABCDh
.code
main PROC
mov ecx, 4
shift right:
    shr BYTE PTR [num+3], 1
    rcr BYTE PTR [num+2], 1
    rcr BYTE PTR [num+1], 1
    rcr BYTE PTR [num], 1
    loop shiftright:

mov num, 2345ABCDh
mov ecx, 4

shiftleft:
  shl BYTE PTR [num], 1
  rcl BYTE PTR [num+1], 1
  rcl BYTE PTR [num+2], 1
  rcl BYTE PTR [num+3], 1
loop shiftleft:

exit
main ENDP
END main
```

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Q3)

```
INCLUDE Irvine32.inc
.data
num DWORD 2345ABCDh
.code
main PROC
mov ecx, 2
shiftright:
 shr WORD PTR [num+2], 1
 rcr WORD PTR [num], 1
 loop shiftright

 mov num, 2345ABCDh
 mov ecx, 2

 shiftleft:
   shl WORD PTR [num], 1
   rcl WORD PTR [num+2], 1
   loop shiftleft

  exit
main ENDP
END main
```


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Q4)

```
INCLUDE Irvine32.inc
.data
BSeconds BYTE ?
BMinutes BYTE ?
.code
main PROC
mov dx, 0101100111101010b
;(a) seconds : bits 0-4
mov ax, dx
and ax, 1Fh
mov BSeconds, al

;(b) Minutes : bits 5-10
mov ax, dx
shr ax, 5
and ax, 3Fh
mov BMinutes, al

movzx eax, BSeconds
call WriteDec
call crlf
movzx eax, BMinutes
call WriteDec
call crlf
exit
main ENDP
END main
```

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(Q5) dx = 9A3Fh

ax = 7F80h

; step 1

ax = 7F80h + 90EAh = 106Ah

CF = 1

; step 2

dx = 9A3F + 1 + CF(1)

dx = 9A41h

CF = 1

step 3

ax = 106A + 8120 = 918Ah ; CF = 0

step 4

dx = 9A41 + 0FFFFh = 9A40

CF = 1

dx = 9A40h

ax = 918Ah

CF = 1

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Q6)

a) test ax, ax

b) test bx, 1

c) test dx, dx

d) test dx, 8000h

e) test bl, bl

Q7) dx = 5

cf = 1

dx = dx + ax + cf

dx = 5 + 10 + 1 = 0016h

Q8) Errors:

1) mov esi, val1 and mov edi, val2 load the value stored in the variable, not it's address. since val1 is a QWORD, this is also a size mismatch. They must be OFFSET val1 and OFFSET val2.

2. mov BYTE PTR [esi], al overwrites val1 instead of storing into result. The result needs its own pointer.

3. dec esi / dec edi go the wrong way. X86 is little endian, so the least significant byte sits at the lowest address. start there and move up with inc.

4. mov cx, 8 sets only the low 16 bits of ECX, but loop in 32 bit mode uses all of ECX. Use mov ecx, 8.

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5. There is no program exit. Add exit before

main ENDP.

Corrected Code:

```
INCLUDE Irvine32.inc
.data
val1 QWORD 2040300436204 7A1h
val2 QWORD 0552103 04A2630B2h
result QWORD 0
.code
main PROC
mov ecx, 8
mov esi, OFFSET val1
mov edi, OFFSET val2
mov ebx, OFFSET result
clc
top:
mov al, BYTE PTR [esi]
add al, BYTE PTR [edi]
mov BYTE PTR [ebx], al
inc esi
inc edi
inc ebx
loop top

exit
main ENDP
END Main
```

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Q9)

```
INCLUDE Irvine32.inc
.data
marks DWORD ?
msg1 BYTE "Enter your marks (0-100): ",0
gradeA BYTE "Grade: A",0
gradeB BYTE "Grade: B",0
gradeC BYTE "Grade: C",0
gradeF BYTE "Grade: F",0
.code
main PROC
mov edx, OFFSET msg1
call WriteString

call ReadDec
mov marks, eax

cmp eax, 80
jae is_A
cmp eax, 60
jae is_B
cmp eax, 40
jae is_C
jmp is_F

is_A:
mov edx, OFFSET gradeA
call Writestring
jmp done
```

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```
is_B:
mov edx, OFFSET gradeB
call WriteString
jmp done

is_C:
mov edx, OFFSET gradeC
call WriteString
jmp done

is_F:
mov edx, OFFSET gradeF*
call WriteString
jmp done

done:
  call crlf
  exit
main ENDP
END main
```

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Q10)

Instruction	EAX	EBX	Stack (t → b)
mov eax, 3	3	?	empty
mov ebx, 7	3	7	empty
push eax	3	7	3
push ebx	3	7	7, 3
pop eax	7	7	3
push 10	7	7	10, 3
pop ebx	7	10	3
pop eax	3	10	empty

Eax = 3

Ebx = 10

stack is empty ~~dirty~~ as there were three pushes and three pops.

Q11)

```
INCLUDE Irvine32.inc
.data
num1 SDWORD ?
num2 SDWORD ?
num3 SDWORD ?
num4 SDWORD ?
result SDWORD ?
.code
main PROC
mov eax, num1
cmp eax, num2
jle checkElseIf
```

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```
mov eax, num3
cmp eax, num4
je checkElseIf

mov eax, num1
add eax, num3
mov result, eax
jmp done

checkElseIf:
mov eax, num2
cmp eax, num4
jne elseBlock:

sub eax, num1
mov result, eax
jmp done

elseBlock:
mov result, 0

done:
 exit
main ENDP
END main
```


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Q12)

```
INCLUDE Irvine32.inc
.data
op1 SDWORD ?
op2 SDWORD ?
x   SDWORD ?
y   SDWORD ?
z   SDWORD ?
.code
main PROC

whileStart:
mov eax, op1
cmp eax, op2
jg endWhile

cmp eax, x
jL elseBlock
cmp eax, y
jg elseBlock
add z, 10
jmp decrement

elseBlock:
sub z, 10
decrement:
dec op1
jmp whileStart
```


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```
endWhile:
exit
main ENDP
END main
```